Building a Precise Video Language with Human-AI Oversight

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https://arxiv.org/abs/2604.21718

Research Questions

- How can we develop a structured specification for precise video captioning that addresses the inconsistencies in existing video-text datasets?
- Can a human-AI oversight framework where experts critique model-generated captions improve annotation quality and model performance?
- How do explicit preference and critique signals from human oversight enhance post-training methods for video-language models?

Core Idea

Existing video captioning datasets lack clear annotation guidelines, leading to inconsistent descriptions that cause models to hallucinate or miss key visual details. The authors address this by creating a structured specification with professional video creators and implementing CHAI (Critique-based Human-AI Oversight), where models generate initial captions and human experts focus on critiquing rather than writing from scratch. This division of labor leverages each party's strengths: models handle fluent text generation while humans focus on visual verification and correction. The resulting critiques and preference pairs provide rich supervision signals for improving models through SFT, DPO, and inference-time scaling.

Key Figure

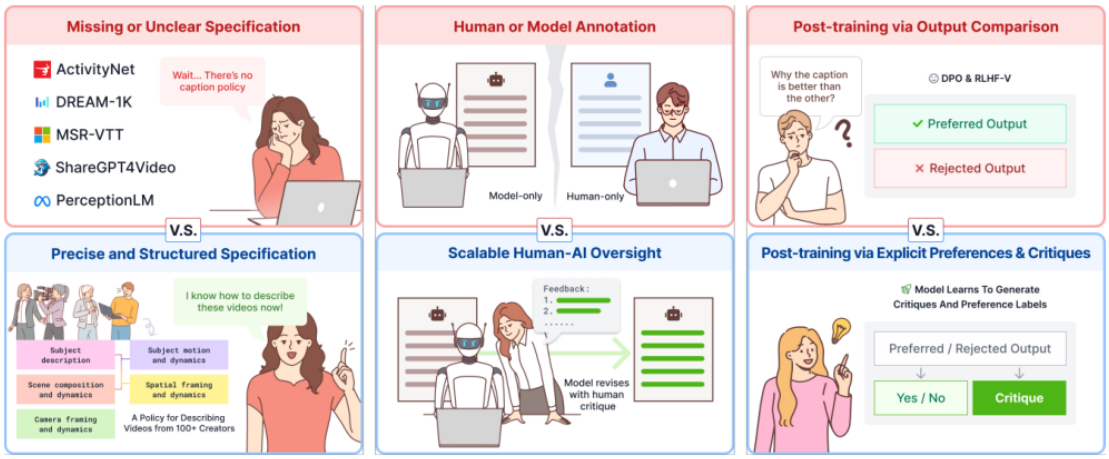

Figure 1. **Recipe for precise video language.** We present a recipe for producing high-quality video captions (blue, bottom row) and compare it with prior work (red, top row). (1) While prior video–text datasets lack a clear policy to teach annotators and models *what* and *how* to describe, we develop a **structured specification** with professional video creators to collect descriptions that support comprehensive video understanding. (2) Instead of relying solely on human or model annotators, we introduce a scalable **human–AI oversight** framework where humans critique model-generated captions with LLM-assisted writing to ensure annotation quality. (3) Our critique-based human-AI (CHAI) oversight framework naturally produces preference pairs and natural-language critiques for **post-training** models, leading to better performance via SFT, DPO, and inference-time scaling. Figure 5 shows that video generators fine-tuned on our captions follow detailed prompts with professional-level control over camera dynamics, shot composition, and cinematography. We release our recipe, data, and models to support future research in fine-grained video understanding and generation.

Figure 1: Shows the complete recipe for precise video language, comparing their structured approach (blue) with prior unstructured methods (red) across specification, oversight, and post-training phases.

Limitations

- Requires expert annotators with professional video creation experience, limiting scalability
- Quality control depends heavily on critique precision, recall, and constructiveness which may be difficult to maintain consistently
- Framework focuses on professional video content and may not generalize well to casual user-generated content
- Initial model generation still prone to hallucinations that require human correction, creating dependency on human oversight